



Image Data Types

There is a wide variety of medical images. Because of this medical images are stored in a variety of ways. When working with medical images, you'll find it useful to have knowledge of the various data types used to store images so you can work with them effectively.

Navigate to the Module 1 folder in the course files and work through the **ImageDataTypes.mlx** script. You'll work with the X-ray image you saw in the previous video and learn how to convert between different data types.

While we recommend you run and experiment with the code in MALTB, you can read a static version below. The interactive tools are not present in the static version.

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1. `imshow(img, [])` - Use the minimum and maximum values in the image as the display limits. This is a quick and fast way to view the data.
2. `imshow(img, [low high])` - Manually specify the display limits. This is typically done to improve the visibility of a specific structure in the image.

Adding a colorbar to the image is a good way to communicate the display range to others.

```
% Create a tiled layout to view the images side-by-side
figure
tiledlayout(1,2)
nexttile

% Approach 1: Automatically use the minimum and maximum values in the image
imshow(img, [])
title("display range = data limits")
colorbar

% Approach 3: manually set a custom display range
nexttile
imshow(img, [800 3000])
title("Custom display range")
colorbar
```

Adjusting the Intensity Values

Adjusting the display range does not change any of the intensity values, only how bright they are when displayed. Adjusting the actual data is often necessary when working with large datasets or images from different machines. Modifying your image data to occupy the same range of values enables you to compare images.

While there are many types of adjustments you can make, here we focus on two of the most common:

1. Linearly scaling the intensity values to use the full range of the data type.
2. Converting between data types.

Scaling the intensity values

The intensity values of the back X-ray are 0 - 4095, but the `uint16` data type has a range of 0 - 65,535. Use the `imadjust` function to adjust the intensity values to use the full range of the `uint16` data type.

```
imgAdj = imadjust(img);
```

You can then view the adjusted image using `imshow` without needed the second input argument. The colorbar shows that the image is now using the full data range of the `uint16` data type.

```
figure
imshow(imgAdj)
colorbar
```